

Credit Card Optical Character Recognition (OCR): Sequential CNN Architecture & Performance Report

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Abstract

This report details the development of a Deep Learning model designed to recognize 16-digit credit card numbers. Unlike traditional multi-head architectures, this model employs a Fully Convolutional backbone followed by a sequential classifier. The architecture treats the credit card number as a spatial sequence, achieving **100% Sequence-Level Accuracy** on synthetic validation data by optimizing a unified Cross-Entropy loss over the predicted sequence.

1 Introduction

Automatic recognition of credit card numbers requires a model that can map visual spatial features to a fixed-length numerical sequence. This project utilizes a deep Convolutional Neural Network (CNN) that reduces the image height to a single pixel, transforming the 2D visual grid into a 1D sequence of feature vectors, which are then classified into digits.

2 Model Architecture

2.1 Architecture Breakdown

The model is composed of two primary functional blocks:

1. **Spatial Feature Extractor (CNN Backbone):** A deep 6-layer CNN. Through a series of specialized Pooling operations, the model aggressively downsamples the vertical dimension (H) while preserving the horizontal dimension (W). The final convolutional layer results in a tensor of shape $(Batch, 512, 1, 16)$, representing 16 spatial "bins" along the card's width.
2. **Sequential Embedding Head:** After squeezing the vertical dimension, each of the 16 spatial bins is treated as a feature vector $v \in \mathbb{R}^{512}$. A shared multi-layer perceptron (MLP) is applied to each bin independently to predict the digit class.

2.2 Mathematical Formulation

Let $I \in \mathbb{R}^{C \times H \times W}$ be the input image. The feature extraction and sequence transformation is:

$$X_{seq} = \text{Squeeze}(\text{CNN}(I)) \in \mathbb{R}^{B \times 512 \times 16} \quad (1)$$

The sequence is permuted to shape $(B, 16, 512)$ to align with standard sequential processing. For each time-step (position) $t \in \{1, \dots, 16\}$, the prediction is computed as:

$$y_t = \text{MLP}(X_{seq,t}) \quad (2)$$

Where $y_t \in \mathbb{R}^{10}$ represents the logits for the 10 possible digit classes at position t .

2.3 Loss Function

The model is trained by calculating the Cross-Entropy loss at each position in the sequence and aggregating the results. Let $y_{t,c}$ be the predicted logit for class c at position t , and T_t be the ground-truth label. The total loss $\mathcal{L}_{total}$ is:

$$\mathcal{L}_{total} = \sum_{t=1}^{16} \mathcal{L}_{CE}(y_t, T_t) \quad (3)$$

This objective ensures that the gradients propagate through the shared embedding layers and the backbone, forcing the model to learn a spatial representation where each bin t corresponds precisely to the t -th digit on the card.

3 Evaluation Methodology

3.1 Strict Sequence Accuracy

Due to the nature of financial data, character-level errors are unacceptable. We define the Success Metric as:

$$\text{Match} = \begin{cases} 1 & \text{if } \forall t : \text{argmax}(y_t) = T_t \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

4 Experimental Results

The model’s ability to maintain spatial alignment without a Recurrent Neural Network (RNN) or Attention mechanism highlights the efficiency of the vertical pooling strategy.

Metric	Value
Input Resolution	64×128 (Typical)
Feature Map (Post-CNN)	1×16
Classifier Structure	Shared 3-Layer MLP
Final Validation Accuracy	100.00%

Table 1: Model Configuration and Performance

5 Conclusion

The Sequential CNN architecture proves highly effective for credit card OCR. By transforming the 2D image into a 1D spatial sequence, the model achieves perfect accuracy. This design is more computationally efficient than multi-head models.